Step 1 — Data Warehouse Modelling

Mexora Analytics — Justification Document

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1. Standard Business Queries (Top-Down Method)

R1 — Revenue Analysis

Analyse → Revenue (montant TTC)

En fonction de → Region, product category, quarter

Pour → Delivered orders only (statut = 'livré')

R2 — Product Performance

Analyse → Quantity sold and revenue TTC

En fonction de → Product, month, customer segment

Pour → Year 2024

R3 — Customer Basket Analysis

Analyse → Average basket size

En fonction de → Customer segment (Gold/Silver/Bronze)

Pour → All delivered orders

R4 — Return Rate

Analyse → Return rate percentage

En fonction de → Product category, product name

Pour → All orders (livré + retourné)

R5 — Ramadan Effect

Analyse → Daily average revenue

En fonction de → Product category, Ramadan period flag

Pour → Food category, years 2022-2024

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2. Granularity Justification

Chosen granularity: one row = one order line

This means each row in FAIT_VENTES represents a single product ordered in a single transaction by a single customer on a specific date delivered to a specific city by a specific delivery driver.

Justification: The CEO's questions require product-level analysis (top products, return rate by product) and customer-level analysis (segment basket size). A coarser granularity such as daily totals would prevent us from answering these questions. The order line is the finest grain available from the source system and enables all required analyses.

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3. Measures and Aggregability

Measure	Type	Justification
quantite_vendue	ADDITIVE	Can be summed across all dimensions — total units sold

| montant_ht | ADDITIVE | Revenue excluding VAT — meaningful at any granularity |

| montant_ttc | ADDITIVE | Revenue including VAT 20% — primary KPI for CEO |

| cout_livraison | ADDITIVE | Delivery cost — summable across orders and regions |

| delai_livraison_jours | SEMI-ADDITIVE | Average is meaningful, but SUM is not — a total of delivery days has no business meaning |

| remise_pct | NON-ADDITIVE | A percentage — summing discount rates produces meaningless results. Must be recalculated as weighted average |

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4. Slowly Changing Dimensions (SCD)

SCD Case 1 — Product Category Change (Type 2)

Scenario: The iPhone 16 product was initially classified under "Phones" but was recategorised to "Smartphones" in March 2024.

Chosen type: SCD Type 2

Justification: We want sales prior to March 2024 to remain classified under "Phones" for historical consistency. A Type 2 approach creates a new row with date_debut = March 2024 and closes the old row with date_fin = February 2024. This preserves historical accuracy for trend analysis across periods.

Implementation: dim_produit has date_debut, date_fin and est_actif columns. The fact table always joins to the product record that was active on the order date.

SCD Case 2 — Customer Segment Change (Type 1)

Scenario: A customer moves from Bronze to Gold segment as their 12-month revenue crosses the 15,000 MAD threshold.

Chosen type: SCD Type 1

Justification: For customer segmentation, we want the current segment to be used for all analysis. The CEO's question "which segment has the highest basket?" refers to current segmentation. Tracking historical segment changes would add complexity without analytical value for this use case. A Type 1 overwrite is appropriate — the old value is simply replaced with the new segment.

Implementation: The segment_client column in dim_client is recalculated on every ETL run based on the last 12 months of delivered orders. No history is preserved.

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5. Star Schema Summary

The schema follows a classic star design with:
- 1 central fact table: FAIT_VENTES (49,567 rows)
- 5 surrounding dimension tables

- All joins use integer surrogate keys for performance
- Indexes on all foreign key columns in FAIT_VENTES
- 3 materialized views in reporting_mexora schema for dashboard performance